



Course Title: Microcomputer Application I
Course Code: CSE-2402

Topic: Internet Topics
(Browsing Concepts, Searching in the web, E-mail)

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Course Agenda



01

Browsing Concepts

How web browsers work, URL anatomy, HTTP/HTTPS, cookies & cache



02

Searching the Web

Search engines, Boolean operators, advanced search techniques & tips



03

E-mail

Email anatomy, protocols (SMTP/IMAP/POP3), security & best practices



04

Practice Questions

Multiple-choice, true/false, and short-answer review questions



05

Lab Task

Hands-on browser, search, and email exercises with deliverables

SECTION 1

Browsing Concepts

Understanding how web browsers work,
URL structure, HTTP protocols, and browser security.



How a Web Browser Works



1. User types URL

2. DNS Lookup

3. HTTP Request

4. Server Response

5. Render Page

DNS Lookup

Translates domain name (e.g. google.com) into an IP address (e.g. 142.250.80.78) via Domain Name System servers.

HTTP Request

Browser sends a GET/POST request to the web server including headers (browser type, accepted formats, cookies).

Server Response

Server replies with status code (200 OK, 404 Not Found, 301 Redirect) plus HTML, CSS, JS, or media files.

Rendering Engine

The browser parses HTML → builds DOM tree → applies CSS → executes JavaScript → paints the visual page.

Anatomy of a URL



 **https://** **www.university.edu** **/library/search** **?q=internet&year=2024**

Protocol (https://)

Specifies the communication protocol. HTTPS adds encryption via SSL/TLS.

Domain Name (www.university.edu)

Human-readable address. Subdomains (www), second-level domain, and TLD (.edu).

Path (/library/search)

Directory or file location on the server. Represents the resource being requested.

Query String (?q=internet&year=2024)

Parameters passed to the server. Starts with '?', separated by '&'.

HTTP vs HTTPS & Key Browser Features



HTTP

- ✗ HyperText Transfer Protocol
- ✗ Data sent in plain text (unencrypted)
- ✗ Vulnerable to eavesdropping & interception
- ✗ No server identity verification

URL starts with: http://

⚠ Not safe for sensitive data (passwords, banking)

Port: 80 (default)

VS

HTTPS

- ✓ HTTP + Secure (SSL/TLS encryption)
- ✓ Data encrypted during transmission
- ✓ Protects against man-in-the-middle attacks
- ✓ Server identity verified via certificates

URL starts with: https://

✓ Required for login, banking, e-commerce

Port: 443 (default)

Browser Storage: Cookies, Cache & History



Cookies

- ▶ Small text files stored on your device by websites
- ▶ Store session data, preferences, login tokens
- ▶ Types: Session cookies (expire on close), Persistent cookies
- ▶ Third-party cookies used for ad tracking
- ▶ Example: Remembering you're logged into Gmail

Browser Cache

- ▶ Stores copies of web resources locally (images, CSS, JS)
- ▶ Speeds up page loading on repeat visits
- ▶ Reduces server bandwidth usage
- ▶ Ctrl+F5 = hard reload bypassing cache
- ▶ Example: Logo images load instantly on second visit

Browsing History

- ▶ Records URLs and timestamps of visited pages
- ▶ Used for autocomplete suggestions in address bar
- ▶ Privacy concern: accessible to other users of same device
- ▶ Incognito/Private mode does NOT save history locally
- ▶ Example: Ctrl+H opens history in most browsers

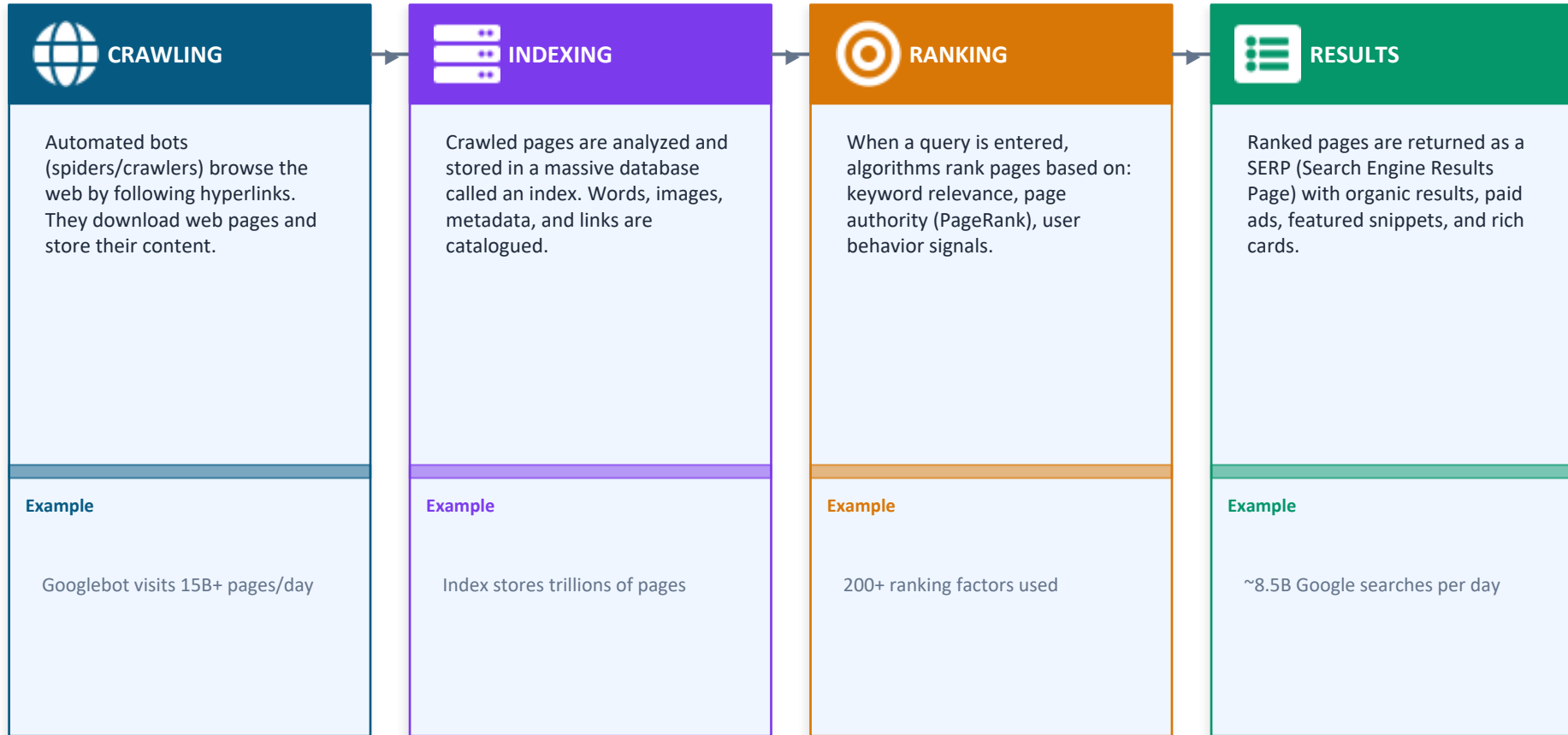
SECTION 2

Searching the Web

Search engine mechanics, Boolean operators,
advanced search techniques and strategies.



How Search Engines Work



Advanced Search Operators & Techniques



OPERATOR	EXAMPLE	DESCRIPTION
"exact phrase"	"climate change effects"	Search for exact phrase
AND / OR	python AND tutorial	Boolean logic operators
NOT or -	jaguar -car (exclude car)	Exclude a word from results
site:	site:bbc.com news	Search within a specific site
filetype:	filetype:pdf research	Find specific file types
intitle:	intitle:machine learning	Search only in page titles
related:	related:wikipedia.org	Find similar websites
define:	define:algorithm	Get dictionary definition

SECTION 3

E-mail Fundamentals

Email architecture, anatomy, communication protocols,
and security best practices.



Anatomy of an E-mail



New Message

From: prof.ali@university.edu

To: student@gmail.com

CC: department@university.edu

Subject: Assignment 3 - Due Friday

Dear Student,

Please find attached the rubric for Assignment 3. Submission is due this Friday by 11:59 PM via the LMS portal.

Best regards,
Prof. Ali

From:

To:

CC:

Subject:

Brief summary of email purpose

Body:

Main message content. Can include HTML formatting, links, images

Email Protocols: SMTP, IMAP & POP3



SMTP

Simple Mail Transfer Protocol

Port 587 (TLS) / 465 (SSL) • Sending protocol

- ▶ Used to **SEND** emails from client to server or between servers
- ▶ One-way: only sends, cannot receive or retrieve messages
- ▶ Works with IMAP/POP3 for complete email solution
- ▶ Authentication required to prevent spam relay

IMAP

Internet Message Access Protocol

Port 993 (SSL/TLS) • Receiving protocol (preferred)

- ▶ Emails stay on the **SERVER** — access from multiple devices
- ▶ Changes sync across all devices in real time
- ▶ Ideal for mobile + desktop use (Gmail, Outlook)
- ▶ Supports server-side folders and search

POP3

Post Office Protocol v3

Port 995 (SSL/TLS) • Receiving protocol (older)

- ▶ Downloads emails **TO** your device, then deletes from server
- ▶ Works offline — no internet needed after download
- ▶ Single device only — no multi-device sync
- ▶ Still used in legacy systems and ISP email

E-mail Security & Best Practices



⚠ Common Email Threats

⊖ Phishing

Fake emails impersonating trusted organizations to steal credentials or install malware. Often has urgent language and suspicious links.

⊖ Spam

Unsolicited bulk email. Can contain malware, ads, or scam offers. Spam filters use ML to block these automatically.

⊖ Email Spoofing

Forging the 'From' address to appear as a trusted sender. Used in phishing and social engineering attacks.

⊖ Malware Attachments

Files (.exe, .docm, .zip) containing viruses, ransomware, or trojans. Never open unexpected attachments.

✓ Best Practices

✓ Use strong, unique passwords

Minimum 12 characters, mix of letters, numbers, and symbols. Use a password manager.

✓ Enable Two-Factor Authentication

2FA adds a second verification step (SMS, authenticator app) making accounts much harder to breach.

✓ Verify sender before acting

Check the full email address, not just the display name. Hover over links before clicking.

✓ Keep software updated

Use up-to-date email clients and operating systems to patch security vulnerabilities.

✓ Use BCC for bulk emails

Blind Carbon Copy protects recipient privacy when sending to multiple people.



Practice Questions — Multiple Choice



Q1. What does 'DNS' stand for and what is its primary function?

- A) Data Network System — manages internet bandwidth
- B) Domain Name System — translates domain names to IP addresses ✓**
- C) Dynamic Network Security — encrypts web traffic
- D) Digital Node Server — stores website files

Q2. Which HTTP status code indicates 'Page Not Found'?

- A) 200 OK
- B) 301 Moved Permanently
- C) 404 Not Found ✓**
- D) 500 Internal Server Error

Q3. Which search operator would you use to find only PDF documents about nutrition?

- A) nutrition.pdf
- B) filetype:pdf nutrition ✓**
- C) type=pdf nutrition
- D) pdf:nutrition

Q4. What is the main difference between IMAP and POP3?

- A) IMAP is used for sending emails; POP3 for receiving
- B) IMAP keeps emails on the server; POP3 downloads and removes them ✓**
- C) IMAP is faster; POP3 is more secure
- D) They are identical — just different versions of the same protocol

Practice Questions — True / False & Short Answer



True / False

HTTPS encrypts data between your browser and the server.

TRUE

Cookies are deleted automatically when you close a browser.

FALSE

(Only session cookies; persistent cookies remain)

The 'CC' field in an email means the recipient will receive a carbon copy.

TRUE

POP3 is better than IMAP for accessing email on multiple devices.

FALSE

(IMAP is preferred for multi-device sync)

Using site:edu in Google limits results to educational websites.

TRUE

Short Answer Questions

Q1. Explain the difference between the browser cache and cookies.
Give one example of each.

[2 marks]

Q2. Describe the steps that occur when you type 'www.google.com' into a browser and press Enter.

[3 marks]

Q3. A student wants to search for journal articles only from IEEE.org about machine learning. Write the search query they should use.

[2 marks]

Q4. Explain why HTTPS is important for online banking. What would happen if HTTP were used instead?

[3 marks]



Lab Task — Hands-On Internet Activities



Lab 1

Browser Investigation

Steps:

1. Open your browser's Developer Tools (F12 → Network tab)
2. Visit <http://www.example.com> and observe the redirect to HTTPS
3. Inspect: HTTP method, status code, response headers, content-type
4. Check cookies set by a site of your choice (Application → Cookies)
5. Clear cache (Ctrl+Shift+Delete) and compare page load time before/after



Deliverable:

Screenshot of Network tab + table: URL, Method, Status Code, Time



Lab 2

Advanced Web Search

Steps:

1. Search for: "machine learning" filetype:pdf site:edu
2. Record the number of results and list 3 relevant findings
3. Use Boolean: AI AND ethics NOT robot — note result count
4. Find the current weather using: weather Dhaka in Google
5. Use Google Scholar to find a peer-reviewed article on cybersecurity



Deliverable:

Report: search queries used, result counts, and one citation in APA format



Lab 3

E-mail Configuration & Practice

Steps:

1. Configure a mail client (Thunderbird/Outlook) with IMAP settings
2. Send an email with: CC to classmate, Subject, attachment (.pdf)
3. Reply-to-all on a class group email thread
4. Identify 3 signs of phishing in a sample email provided by instructor
5. Enable 2-Factor Authentication on your university email account



Deliverable:

Screenshot of sent email + written analysis of phishing sample (150 words)

Summary & Key Takeaways



Browsing Concepts

- ◆ Browsers: DNS → Request → Response → Render
- ◆ HTTPS encrypts data; HTTP is insecure
- ◆ URLs have 4 parts: Protocol, Domain, Path, Query
- ◆ Cookies persist sessions; Cache speeds loading



Searching the Web

- ◆ Use "quotes" for exact phrases
- ◆ filetype: and site: narrow results significantly
- ◆ Boolean (AND/OR/NOT) controls result logic
- ◆ Evaluate sources: check domain, date, author



E-mail

- ◆ SMTP sends; IMAP/POP3 receives email
- ◆ IMAP syncs across devices; POP3 downloads locally
- ◆ Email fields: From, To, CC, BCC, Subject, Body
- ◆ Use 2FA; never click suspicious links or attachments

Thank You

Any Questions?